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FARMER PARTICIPATORY CROP IMPROVEMENT. IV. THE SPREAD AND IMPACT OF A RICE VARIETY IDENTIFIED BY PARTICIPATORY VARIETAL SELECTION

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SUMMARY

Participatory varietal selection in a development project in western India showed that the rice (*Oryza sativa*) variety Kalinga III was highly preferred by farmers. The spatial diffusion of this variety from three villages (two project and one non-project) was studied. Seed of Kalinga III had spread from the three villages in 1994 to 41 villages by 1996 and is estimated to have reached over 100 widely distributed villages by 1997. Farmer-to-farmer spread was as high from the non-project case study village that received no further seed from the project, possibly because farmers are more likely to spread seed of a new variety to other farmers when they have no assured supply. Project interventions used key villages, informal-sector seed merchants, and Non-Government Organizations in the spread of seed. The project also collaborated with Rajasthan State Agricultural University and Kalinga III has been proposed for release in that state. A financial analysis revealed the very high internal rates of return that are possible from investment in participatory varietal selection.

INTRODUCTION

Participatory Varietal Selection (PVS) can be used to identify acceptable new varieties and thereby overcome the constraints that cause farmers to grow landraces or obsolete cultivars (Joshi and Witcombe, 1996; Witcombe *et al.*, 1996). Varieties of several crops were tested in a PVS programme in the Krishak Bharati Co-operative Ltd (KRIBHCO) Indo-British Rainfed Farming Project (KRIBP). The rice (*Oryza sativa*) variety, Kalinga III, was identified as being highly preferred by farmers (Joshi and Witcombe, 1996).

The success of this and other PVS programmes in identifying preferred varieties is well documented (for example, Sperling and Scheidegger, 1995). It is less well understood how improved varieties diffuse through informal channels following their initial adoption, despite the known importance of farmer-to-farmer spread in disseminating new varieties (for example, Maurya, 1989; Chilver and Suherman, 1994).

Factors affecting the informal diffusion of modern varieties (MVs) and the ability of low-resource farmers to obtain seeds and continue to cultivate them have been reviewed (Cromwell, 1990). These include access to credit and availability of suitable land for the new variety (Bellon and Taylor, 1993); farmers' knowledge that enables seed to be retained effectively from year to year (Grisley and Shamambo, 1993); and factors, such as drought, that lead to crop failure and force the poor to buy seed each year (Sperling and Loevinsohn, 1993).

In this paper a study of seed diffusion of Kalinga III in western India is reported and the financial returns to the PVS programme are estimated.

MATERIALS AND METHODS

The project area

The KRIBP project area (Jones *et al.*, 1993) consists of the adjoining districts of Panchmahals (Gujarat), Jhabua (Madhya Pradesh) and Banswara (Rajasthan). Annual rainfall is 700–1100 mm. Average land holdings are less than 1 ha, and soils are infertile, shallow and on sloping land. Farmers grow mostly maize (*Zea mays*) in the rainy season (kharif) but upland, direct-drilled rice is also widely grown.

Selection of the case study villages

Three primary case study villages were chosen in Rajasthan where rice is more widely grown than in Gujarat or Madhya Pradesh. The project had been active in two of the villages, Bijori and Kompura, since 1992. The other was a non-project village, Gamana, that was given seed of Kalinga III once by the project in 1994. Villages that had received seed from these primary villages were also studied. It was found that 45 villages had received Kalinga III seed by 1996 from farmers in other villages, and a sample of 28 of these villages were visited by researchers to establish the fate of the seed. A random sample was not made because of difficulties in locating all of the villages; a village was visited if directions could be obtained or a guide from the donor village was available. Although this could have led to a bias towards more accessible villages, it did not appear to do so – few of the villages were on a metalled road and several were inaccessible by jeep.

Interview techniques

Semi-structured interviews were undertaken in April to October 1997. After introductions, the first question asked of farmers was 'what rice varieties do you grow?' This was to estimate the number of varieties grown in the village. Such estimates were approximate because genetically dissimilar varieties may share the same name, while the same variety may have different names in different villages. For example, some Rajasthani farmers called the recently introduced Kalinga III 'Kalinga' and others 'Gujarati' because they had received it from farmers in Gujarat and may never have been told its real name. If any new varieties were being cultivated, their names were asked. When a farmer was found to grow

Kalinga III the amount of seed sown in each year of cultivation was determined and details of the distribution of Kalinga III seed to other farmers were obtained. Whenever possible, the easily distinguished seed of Kalinga III was examined to confirm its identity.

Yield data

Yield data for 1994 were obtained from farmers' fields for 58 paired comparisons of Kalinga III with local varieties, using methods described in Joshi and Witcombe (1996). Similar data, using the same methods, were obtained in 1996 for 34 paired comparisons on farmers' fields (26 in Rajasthan, 8 in Gujarat).

Financial analyses

An analysis was undertaken to assess the financial returns to the PVS approach as implemented by the KRIBHCO Project. A financial analysis uses market prices rather than the shadow prices, which reflect the value of resources to the national economy of an economic analysis (Gittinger, 1982).

Financial benefits to farmers were estimated using the yield data, market surveys on grain prices conducted by the project, and the findings from farmer interviews.

In order to estimate the financial returns to PVS it was assumed, from the evidence provided by the diffusion study, that Kalinga III would continue to spread throughout the three districts of the project area and into the adjoining six districts where upland rice is also cultivated by the Bhil and related ethnic groups. The total upland area of these nine districts was estimated by the project, from village surveys and remote sensing, to be 1×10^6 ha, of which 20% (about 200 000 ha) is under rice every year.

Rate of spread from village to village was assumed to be lower than that found in this study to avoid any possibility of overestimating benefits. It was assumed that adoption would follow an S shaped curve (Rogers, 1962; Hägerstrand, 1967) so rates were assumed to slow once half of the villages were growing Kalinga III (Fig. 1). Three different adoption ceilings (30, 45 and 60%) were used, with the highest somewhat lower than the highest found in this study, with two growth rates within villages approximating to those found in this study, that is, four or five years to achieve the ceiling (Fig. 2).

The Financial Internal Rate of Return of PVS was estimated using KRIBP data on project costs and benefit estimates for each of the adoption ceilings and growth rates assumed.

RESULTS

Spread from Kompura

Kalinga III was first tested in Kompura in 1993 in a Farmer-Managed Participatory Research (FAMPAR) trial where farmers were given 1 kg seed to grow under unchanged management conditions (Joshi and Witcombe, 1996).

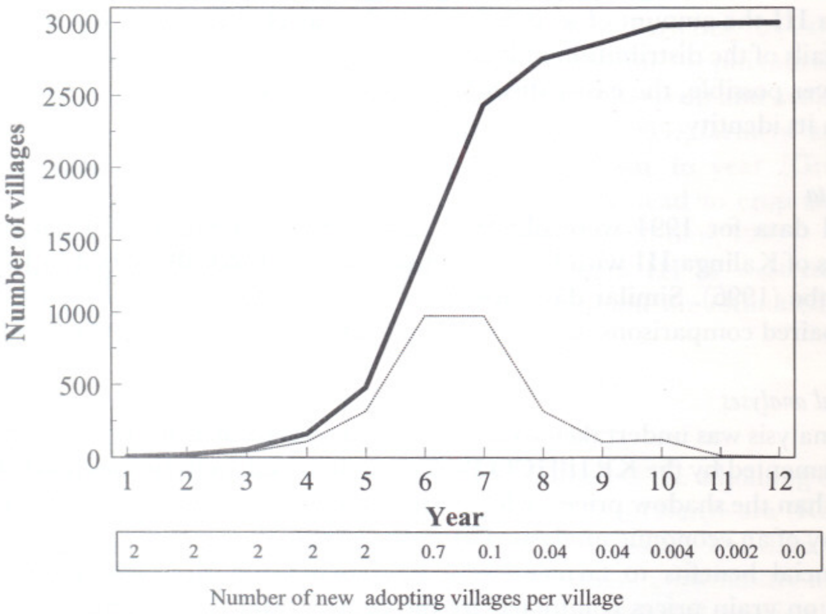

Fig. 1. Predicted adoption curves of Kalinga III in nine districts by number of villages. (Cumulative total of villages solid line, new villages dashed line.) The number of newly adopting villages per village is shown below the graph.

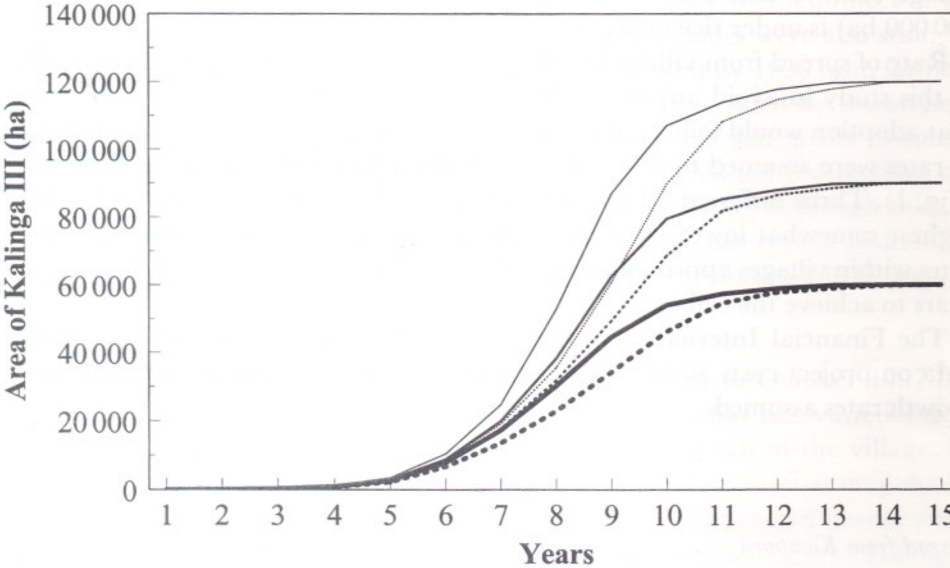

Fig. 2. Predicted adoption curves of Kalinga III in nine districts by hectare. (Dotted lines indicate spread within villages so ceiling is achieved in 5 years, solid lines for ceiling achieved in 4 years. Thick lines 30% adoption ceiling, medium lines 45%, and thin lines 60%.)

Many farmers retained seed from the 1993 harvest for the following sowing, though no spread to other villages was reported. Since 1993, the project has sold seed to Kompura farmers each year, at slightly above the local market rates for rice seed, but on cheap credit through the project-initiated savings and credit group. Project-organized seed pools of Kalinga III have also been established and this made it possible for the village to act as a distributor of seed to other villages (Fig. 3). Early on, Kompura farmers passed seed to farmers in nearby villages (within 5 km) to whom they were related by blood or marriage. Most of the seed was sold for cash though small amounts of seed were donated. In 1995, of the 310 kg distributed to farmers from Kompura, 115 kg was supplied from a seed pool owned by a group in the village. The group got back after harvest the original quantity plus 25%.

Farmers from Kundini Rupa village were impressed with the performance of the crop that was grown in 1996 by two farmers who had obtained seed from Kompura. As a result, in 1997 almost all households obtained seeds and grew Kalinga III. Also in 1997, a *jankar* (a villager trained by the project) was so enthusiastic about Kalinga III that he offered to distribute seed. The project supplied 4 t seed which the *jankar* and other farmers from Kundini Rupa sold in their own village and to friends and relatives in six more villages (Fig. 4).

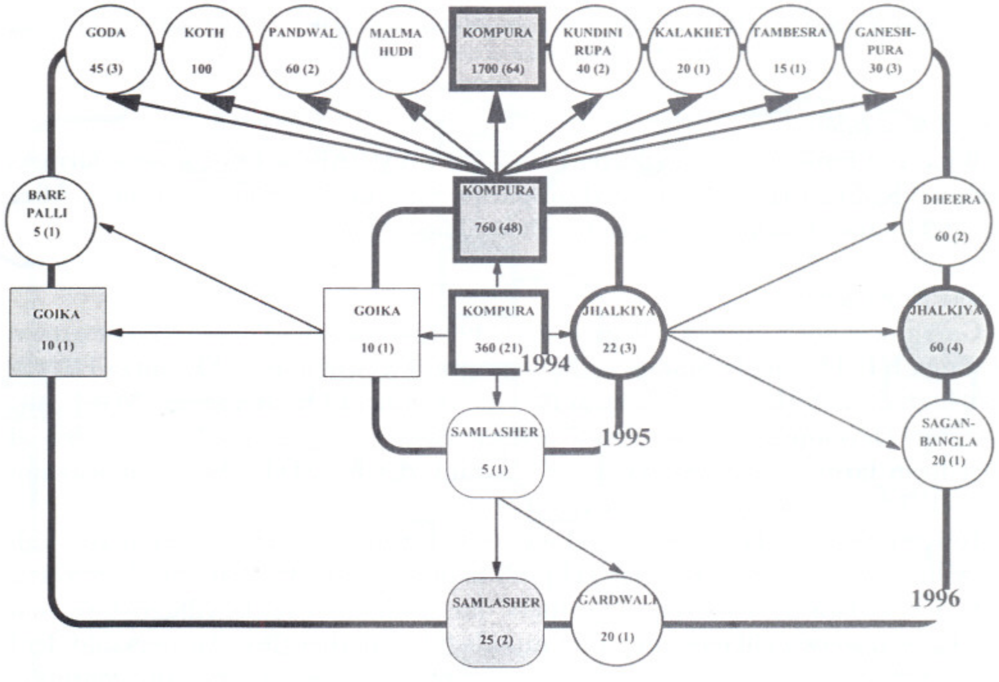

Fig. 3. Spread of Kalinga III from Kompura, Rajasthan, 1994–1996. Villages that occur once are indicated by a circle with a fine-lined border. Each village that occurs more than once has a unique symbol that is shaded for second and third year adoption. Seed distributed from the first harvest is indicated by thin arrows, from the second harvest by thick arrows. Figures refer to kilograms of seed sown and the numbers in parentheses refer to the numbers of households.

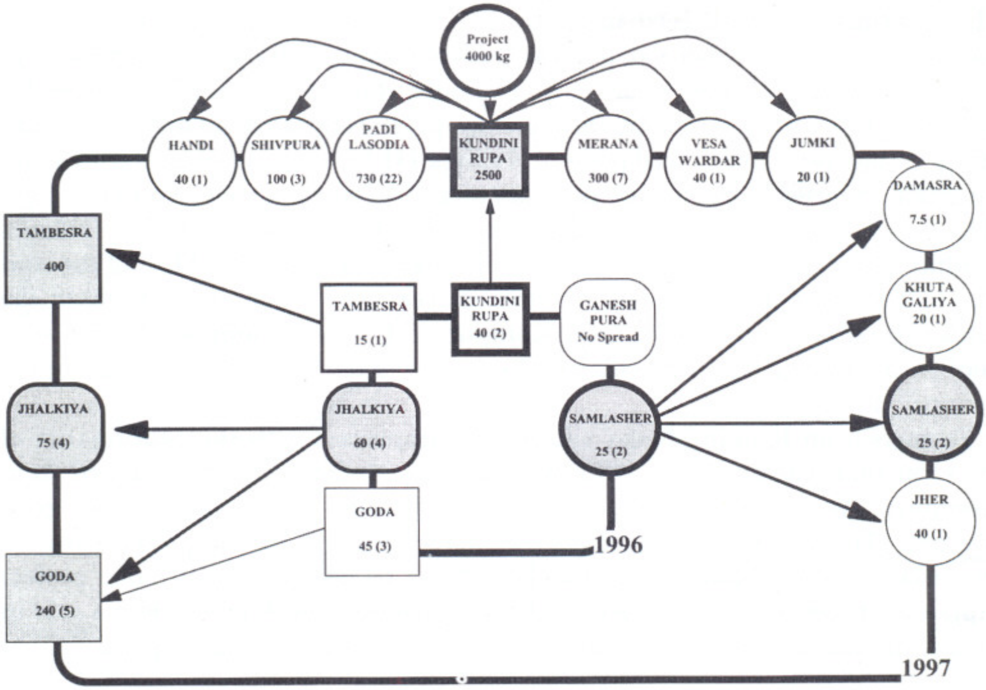

Fig. 4. Spread of Kalinga III, 1996–1997, from a sample of six villages that received seed from Kompura in 1995 or 1996. For explanation of symbols and figures see Fig. 3.

Spread from Bijori

Bijori is a remote project village. Following FAMPAR in 1993, several farmers adopted Kalinga III in 1994. Seed supplies were available from the project. After the 1994 harvest, diffusion began to other villages (Fig. 5).

Spread from Gamana

Gamana is not a project village and, in 1994, farmers had never seen or grown Kalinga III. The single contact with the project was when seed of Kalinga III was distributed in 1994, through a local Non-Government Organization (NGO), the Sadguru Development Trust (SDT). Many of the villages, which later received seed from farmers in Gamana, were also linked with SDT, though it was not actively involved in the seed dissemination.

In 1994, two brothers, whose land is close to the metalled road running through the village, showed their Kalinga III plots to others and, after harvest, there was a high demand for the seed. These brothers, one of whom owns the village store, sold 380 kg seed to two villages at market rates, whilst another three farmers sold, lent or gave 29 kg to a further three villages (Fig. 6). The owner of the store continues to cultivate the variety, has sought to purchase seed directly from KRIBP, and is able to sell it for Rs1 kg^{-1} more than the normal market price. Spread from Monadungar, which received seed in 1995, was very high after the first harvest (Fig. 6).

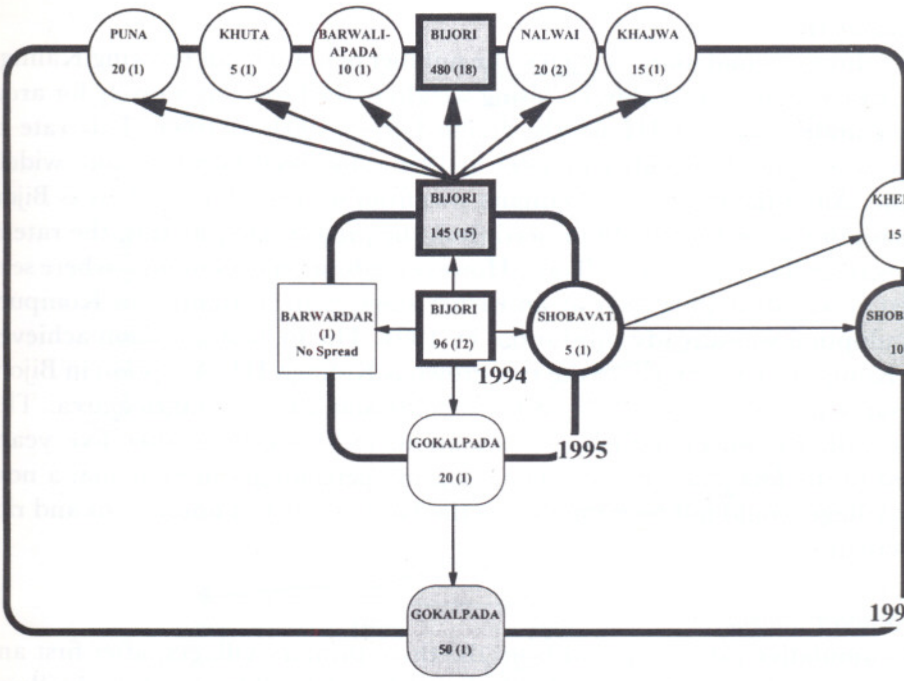

Fig. 5. Spread of Kalinga III from Bijori, Rajasthan, 1994–1996. For explanation of symbols and figures see Fig. 3.

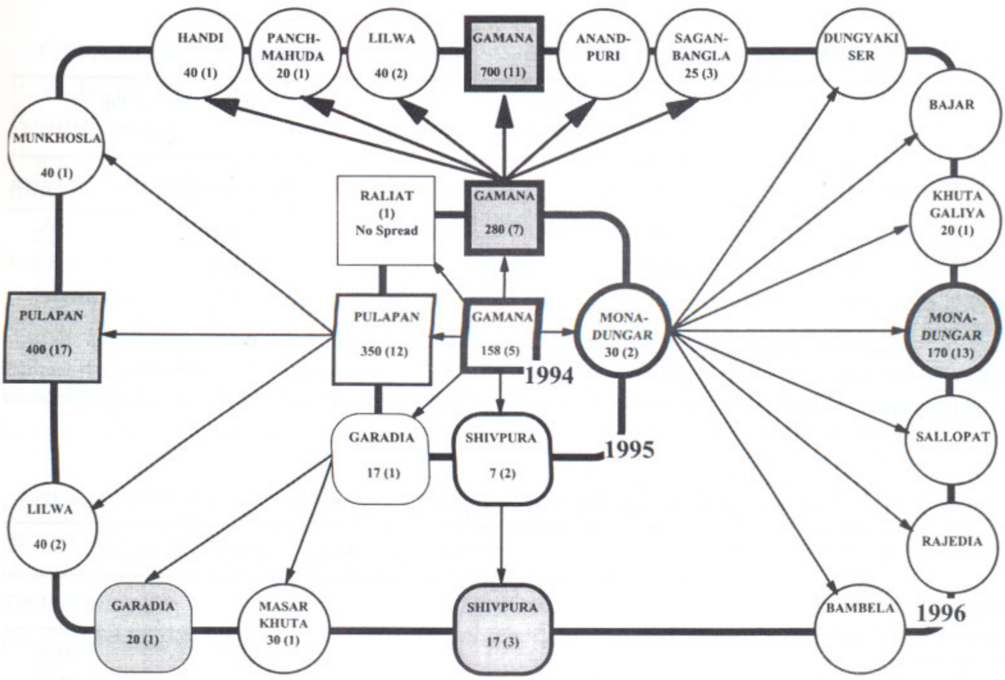

Fig. 6. Spread of Kalinga III from Gamana, Rajasthan, 1994–1996. For explanation of symbols and figures see Fig. 3.

Spread within villages

In the three original source villages, the number of households growing Kalinga III increased each year (Table 1). Using weight of seed sown as a proxy for area, the area under Kalinga III nearly doubled from 1994 to 1995. This rate of increase was somewhat higher in 1995 to 1996, after seed became more widely available. All of the increase in Gamana was from harvested seed whereas Bijori and Kompura were able to obtain seed from the project. Despite this, the rate of spread was similar in all three villages. However, adoption in Gamana, where seed was first received in 1994, was at an earlier stage than in Bijori and Kompura where adoption was already substantial by 1994. The highest adoption achieved was in Kompura where 65% of the rice grown is Kalinga III. Adoption in Bijori, after four years, was about 25% of households and 20% of the rice area. This accords with the lowest estimate of adoption, a 30% ceiling after five years, assumed in the financial analysis. The adoption percentages in Gamana, a non-project village, could not be estimated as the total number of households and rice area were unknown.

Spread between villages

The cumulative rates of spread from the three primary villages, after first and second harvests, were very high (Table 2 and Fig. 3–6). The number of villages increased by a factor of 2.3 to 7.0 (Table 2). Rates of spread were higher after the second harvest than after the first.

Table 1. Rate of spread of Kalinga III within the three study villages by household and area.

	1994–1995 Rate†		1995–1996 Rate†	
Spread within	By household	By area	By household	By area
Kompura	2.3	2.1	1.3	2.2
Bijori	1.2	1.5	1.2	3.3
Gamana	1.4	1.8	1.6	2.5
Overall‡	1.8	1.9	1.3	2.4

† Rate = factor by which household numbers and area increase, for example, 2 is a doubling in number; ‡ represents the overall rate of increase in the number of households and area. It does not equal the mean of the three villages because the larger villages have greater weight.

Table 2. Rate of spread of Kalinga III from the three study villages 1994–1997 by number of villages.

	Rate† of increase in number of villages		
Spread	1994–1995	1995–1996	1996–1997
From first harvest	4.7	2.3	2.5
From second harvest	—	7.0	3.0

† Rate = factor by which village numbers increase, for example, 2 is a doubling in number.

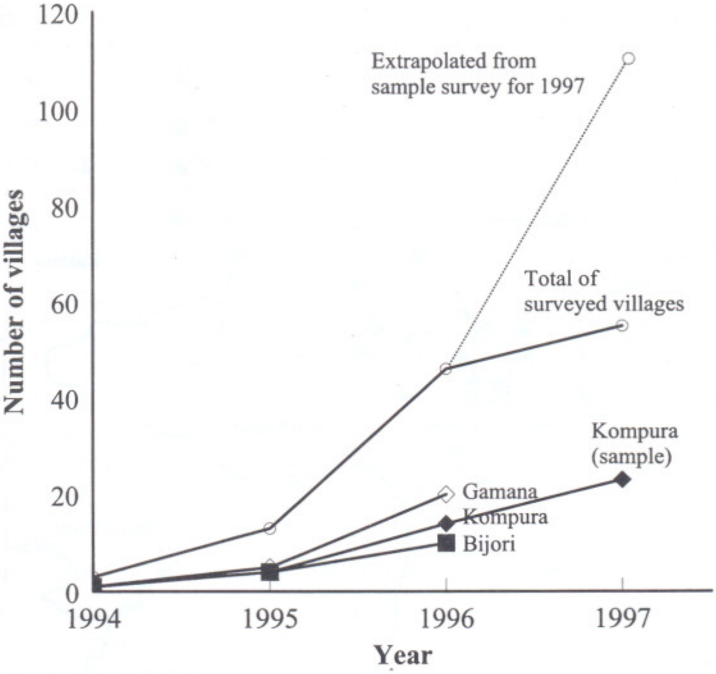

Fig. 7. Documented and extrapolated spread of Kalinga III from 1994 to 1997 from three villages in 1994.

Overall spread from Gamana (to 19 villages by 1996) was at least as high as from Kompura (to 16 villages by 1996), despite the fact that project assistance to Kompura was much higher. Spread from both Kompura and Gamana was higher than from Bijori (to 9 villages by 1996), possibly because Bijori is a more inaccessible village.

All of the villages which received Kalinga III in 1995 from the case study villages were visited by researchers in 1996 and it is known that all of them sowed the seed. Of the 44 villages that received seed in 1996, 28 were visited and all of them had sown the seed. Assuming that the remaining 16 villages also sowed the seed they received and spread seed from the harvest at similar rates to the 28 sampled villages, by 1997 Kalinga III had spread from the three original villages to over 100 surrounding villages (Fig. 7).

Although the increase in the number of villages growing Kalinga III was very high, new villages typically started with only 5–15 kg seed unless there was project intervention. However, the quantities available grew quickly (multiplication rates of over 100 were reported) and demand for seed from other sources was created.

Geographical diffusion pattern

The main pattern of seed diffusion was from farms to relatives or friends in adjacent or nearby villages typically over distances of less than 10 km (Fig. 8). However, there were jumps of over 50 km from Gamana and Kompura as the

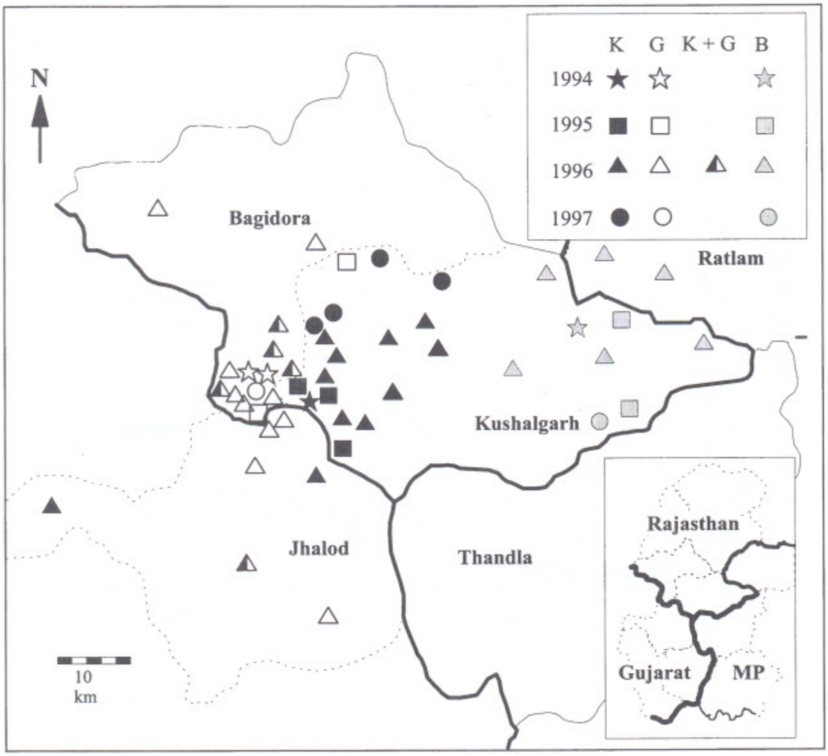

Fig. 8. Map of project area showing all villages known by 1997 to have adopted Kalinga III that originated from the three study villages. K = originating from Kompura, G = originating from Gamana, K + G = originating from both Kompura and Gamana, and B = originating from Bijori.

result of sales from seed merchant farmers and distant family links. Apart from these outliers, there was some difference in the average distance seed spread from the three primary villages, as the spread from Gamana was somewhat less diffuse.

Role of seed merchants

About 14% of the Kalinga III harvest from Kompura was sold to merchants or other farmers. Although it was not possible to know if some of this grain was used for sowing, it seemed likely that it was. Hence, the informal diffusion of Kalinga III was probably underestimated.

It was established from interviews that merchants first paid less for Kalinga III than for local varieties. However, in areas where Kalinga III was becoming known, they bought it at the same price as the local varieties. Farmers sold Kalinga III for a premium amongst themselves and seed merchants also charged a premium.

Although most rice seed sown was saved from the previous harvest (for example, in Kompura, 78% of local rice varieties and 86% of Kalinga III) poorer families were more likely to buy seed each year as they usually consumed or sold all of their harvest.

Role of the project

KRIBP disseminated Kalinga III seed throughout the project area. In 1994, apart from the case study villages, Kalinga III was distributed to 12 other villages. The project is currently working in 75 villages, and by 1996, had sold more than 10 t seed in project villages, had given nearly 2 t seed for FAMPAR trials in new villages and had supplied over 7 t seed to Government Organizations (GOs) and NGOs. In 1997, 16.5 t was sold in project villages, 4.5 t was given in new villages and 3.9 t supplied to GOs and NGOs.

In Kompura village, farmers no longer need seed from the project as they have a functioning seed pool that meets their needs for sowing. However, the project continues to distribute seed to Kompura because of demand from farmers and this village acts as a 'key distributor' of seed to other villages. Similarly, Kundini Rupa village acted as a key distributor when it sold project-supplied seed.

The savings and credit groups in KRIBP villages provide an additional incentive for farmers to buy project-supplied varieties, but this is not an overriding factor in the spread of Kalinga III as nearly all of the secondary villages covered by this study are not project villages.

Biodiversity

Although adoption of Kalinga III has been high, most farmers continued to grow local varieties as well. Kalinga III is suited especially to drier slopes and is not likely to become the only variety of rice grown in any village.

Financial analysis of the benefits to farmers of growing Kalinga III

Comparison of costs and benefits showed there are significant financial advantages to be gained from growing Kalinga III instead of local rice varieties under the same conditions, which partly explains the high acceptability of the variety (Table 3). In the financial analysis, the same grain price was used for new and old varieties to estimate the gross value of production although it was expected that farmers would be able to sell the new variety at a premium.

Financial analysis of participatory varietal selection

The financial analysis was based on the returns to production in Table 3 and this gave an incremental benefit of Rs 2113 ha^{-1} from Kalinga III cultivation compared with local varieties, grown under the same management system (Table 3). A conservative assumption was made that PVS accounted for 33% of the total KRIBHCO farming systems development project costs, including technical assistance consultants and overseas training. It was also assumed that, without the project, Kalinga III would have eventually reached the area so that the benefits would have accrued only for a period of 15 years from first introduction by the project.

Table 3. Comparison of the costs and benefits of growing Kalinga III (K III) compared with local varieties.

Year	Yield ($kg\ ha^{-1}$)			Grain price ($Rs\ kg^{-1}$)			Gross value of production (Rs)		
	K III	Local	Difference	K III	Local	Difference	K III	Local	Difference
1994	1521†	1042†	479 (46%)				5665§	3949	1716
1996	2632‡	1673‡	689 (41%)	4.14	3.27	0.8 (27%)	8852§	6341	2511
Value¶	1942	1358	584 (43%)	3.79	3.79	0 (0%)	7260§	5147	2113

†Joshi and Witcombe (1996), significantly different ($p < 0.001$); ‡project data (see materials and methods), significantly different ($p < 0.001$); §value reduced by additional cost of planting Kalinga III seed instead of local, estimated at a sowing rate of 100 $kg\ ha^{-1}$ to give an added cost of Rs 1 kg^{-1}; ¶the value used in the financial analysis, the mean for yield and net benefit, assumed to be 0 for grain price.

Various assumptions were also made of the maximum proportion of village land on which Kalinga III would be grown, and the time taken to achieve that ceiling in a given village. The financial return to the investment in PVS based on all these assumptions was good, ranging from 47% to 70 % (Table 4).

DISCUSSION

Little is known about how seed of varieties supplied by NGOs diffuses from farmer to farmer after initial adoption (Wiggins and Cromwell, 1995). Although farmer-to-farmer spread has been documented, for example, Maurya (1989) described the spread of Masuli rice from Andhra Pradesh to Orissa, details of actual processes are little known. This may be because of the cost and difficulties of undertaking such studies. Starting from only three case study villages in 1994, Kalinga III had spread to about 100 villages, in an area of several thousand square kilometres, by 1997.

Factors influencing farmer-to-farmer spread

Kalinga III was found, using participatory methods, to have many perceived advantages over local varieties, including high yield, short duration and high grain quality (Joshi and Witcombe, 1996). The financial benefits to farmers of growing Kalinga III instead of local varieties were considerable. These percep-

Table 4. Financial returns to participatory varietal selection.

Time needed to achieve ceiling	Financial internal rate of return (%) for three proportions of rice land ('ceiling') under Kalinga III		
	30% ceiling	45% ceiling	60% ceiling
Five years	47	60	64
Four years	56	63	70

ons and advantages were confirmed by the rapid farmer-to-farmer spread of ;alinga III since 1994. Many studies have highlighted the paramount import-nce of farmers' perceptions in the adoption of new varieties. For example, .desina and Seidi (1995) found that earlier yield, high yield and ease of threshing ·ere the key attributes that determined farmers' decisions to adopt modern ıangrove rice varieties in Guinea Bissau. Key determinants of adoption of rice in ri Lanka were ability to perform under adverse conditions and higher farm gate rice (Wijeratne and Chandrasiri, 1993).

Farmers in non-project villages, such as Gamana and Monadungar, spread .alinga III to many other villages. One reason may have been to provide a seed ›urce in case their own crop failed. For example, Chilver and Suherman (1994) ı Indonesia found that farmers not participating in on-farm trials disseminated ɔtato (*Solanum tuberosum*) seedling tubers to more individuals than farmers who ere involved in the trials. Grisley and Shamambo (1993) found that farmers uickly established an informal community seed pool for beans (*Phaseolus vulgaris*) ⁄. Carioca in Zambia – after the first harvest 41% of farmers had given seed to :latives and neighbours and this fell to 10% after the third harvest.

Seed availability appears to be an important factor in influencing the rate of ›read. The high spread of Kalinga III between villages, that began after the first ɪrvest, increased after the second, when more seed was available. This finding as expected, but no reports were found in the literature of studies of this process. Sperling and Loevinsohn (1993) in Rwanda, found previously untested bean ʾ*haseolus vulgaris*) cultivars did not start to spread until they had been tested in rmers' own fields for three years. FAMPAR trials provide a rapid means of sting new varieties widely over a range of soil types and management systems ıd should accelerate the process of dissemination.

As reported in other studies, spread between villages was on the basis of kinship, ually within the same generation. For example, Chilver and Suherman (1994) und that 40% of potato seedling tuber exchanges in Indonesia were with close mily members, and 20% with friends and neighbours. In the current work a ealth bias towards the better off was found in the early stages of spread, and this ıs been reported elsewhere (Otsuka *et al.*, 1994; Sperling and Loevinsohn, 1993) both traditional and formal diffusion mechanisms.

Kalinga III has jumped large distances, often because a recipient family ember married into a distant village or because of seasonal migration by local rmers to urban centres. Spread has covered a larger area than was found by reen (1987) for rice in Nepal, where communications are physically more fficult and by Tetlay *et al.* (1990), who found that, in three districts of Pakistan, rmer-to-farmer spread of wheat (*Triticum aestivum*) operates primarily at the llage level, or within 5 km of home.

There were clear geographic patterns of diffusion outward from the three case ıdy villages. Huke and James (1969) found a similar pattern for IR8 rice from iginal centres in the Philippines, with a faster spread along a national road. The ean distance of spread, between farms rather than between villages, varied across years from 2.3 to 5.1 km. Such geographical patterns of spread mean that project interventions will have the greatest effect if seed is distributed to widely dispersed villages from which outward spread will occur.

Other methods of dissemination

Cromwell *et al.* (1993) noted that little is known about the impact of seed distribution in the informal sector and that activities by NGOs in this field are generally commercially unrewarding, recovering only 25–50% of the costs involved. Cost recovery on the sale of seed in KRIBP was certainly low in terms of meeting project overheads but the financial analysis demonstrated the large benefit of the activity to farmers. The distribution of several tonnes of seed to NGOs and GOs working in the project or adjoining areas has been a very effective approach in scaling up the outputs from KRIBP's research.

Kalinga III has not yet been released in Rajasthan, Gujarat or Madhya Pradesh, but collaborative research on rice with Rajasthan Agricultural University has resulted in a release proposal for the variety in Rajasthan. Release will mean that the variety qualifies for official subsidies.

In the absence of a well established formal seed sector, many methods have been used by KRIBP to disseminate seed: repeated distribution of seed to key villages (Kompura and Bijori); a single large influx of seed to a roadside village (Kundini Rupa) which had shown great interest in Kalinga III; distribution of seed to a store owner in one village (Gamana), who has since approached the project for more seed for sale; and the establishment of seed pools in project villages (for example, Kompura). Green (1987) regarded the informal seed exchange system as a rarely used asset, and recommended the targeting of certain villages as seed villages, from which improved varieties could spread through the existing informal system. The project has begun to take this further by encouraging farm- and village-level multiplication and distribution systems, a strategy recommended by Wiggins and Cromwell (1995) and reported to be successfully implemented in Cameroon (Moussie and Foaguegue, 1994). Such a system operates in the same way as the principle of 'key distributors' identified by Sperling and Loevinsohn (1993). The use of stores, such as in Gamana and Sagan Bangla, another strategy recommended by Sperling and Loevinsohn (1993), ensures that diffusion is through a wider social spectrum than to the relatives of better-off farmers. Kundini Rupa played an equivalent function to a store when farmers there sold 4 t seed for the project.

Farmer-to-farmer spread versus project interventions

An attempt can be made to separate the impact of the project from informal spread. Once seed had been supplied to Gamana by the project, seed spread on a village-to-village basis was higher from Gamana, a non-project village, than from Kompura or Bijori. Project seed supply increased spread within Kompura and also increased the quantity of seed entering new villages in the first year of adoption, for example, 730 kg to 22 households in Padi Lasodia (Fig. 4) compared with typical quantities of 20–40 kg for one to two households by farmer-to-farmer spread (Fig. 6). Project interventions certainly increase the rate of spread within villages, and can make spread between villages more effective, but in the absence of such interventions farmer-to-farmer spread between villages is an effective but slower process.

Financial analysis

The financial analysis shows that, as a result of its rapid adoption, Kalinga III is producing very high financial benefits. This demonstrates the cost effectiveness of PVS when highly preferred cultivars are identified. This was confirmed by Balogun (1996) who estimated that the 'loss' to farmers in the six upland rice growing states of India, where Kalinga III is suitable for cultivation but has not been released or recommended, amounted to Rs10 702 $\times 10^6$, or £198 $\times 10^6$ at 1996 discounted prices, compared with the likely returns had Kalinga III been released in 1983, the year it was released in Orissa.

As has been found elsewhere, the most important variable determining the rate of return of a new technology is the adoption ceiling determined by the proportion of suitable land that can be occupied by the new variety and the proportion of farmers having such land that are adopters (Bellon and Taylor, 1993; Mazzucato and Ly, 1994). The speed at which adoption occurs has a much less marked effect on the rate of return. The proportion of suitable land is outside the influence of a project but speed of adoption and the proportion of farmers adopting can be increased by carefully planned project-led seed dissemination.

The financial analysis also indicates that the high financial returns to PVS are probably enough to allow private sector enterprises to adopt this approach as a profitable venture provided they have a good reputation with farmers and can produce sufficient quality seed to sell as the demand rises. The application of such a commercial approach is not limited either to marginal areas or to rice. In high potential production systems in Nepal, Kalinga III is spreading rapidly as a result of PVS (K. D. Joshi, personal communication). PVS has also been used successfully in KRIBP to identify varieties of chickpea (*Cicer arietinum*) (Joshi and Witcombe, 1996), maize (*Zea mays*) and black gram (*Vigna mungo*), (Joshi and Witcombe, 1998) and these varieties are also spreading within and outside the project area.

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